

Week 8: Confidence Intervals

Part 2

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Overview

- We will finish the topic of confidence intervals this week
- Topics for this week
 - Sample sizes
 - Confidence Intervals for the mean
 - Robustness of confidence intervals
 - T Distribution

Review: CIs for Proportions

Review: CIs for Proportions

- We are trying to find an estimate for the population proportion p by collecting a sample and computing $\hat{p}$
- We can then build a $(1 - \alpha)\%$ confidence interval around $\hat{p}$ to give us a range of plausible values that p could be.
- The confidence interval is built by taking the estimate $\hat{p}$ and putting limits on either side, determined by the standard error $SE(\hat{p})$ (i.e. SD from the sampling distribution)
- By finding $z_{\alpha/2}$, a critical value from the standard normal distribution, we can determine how many SEs away from $\hat{p}$ to be included in the interval

Review: CIs for Proportions

- We end up with the interval

$$\left(\hat{p} - z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}}, \hat{p} + z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}} \right)$$

- Since we don't know p , we can either use $\hat{p}$ or 0.5 in the standard error term.

Sample Sizes

Sample Size

$$SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

- If we look at the expression for the confidence interval, we basically are working with 3 pieces:
 - the sample proportion $\hat{p}$
 - the critical value $z_{\alpha/2}$ based on confidence level $(1 - \alpha)\%$
 - the sample size n
- We have already discussed how the value of $\hat{p}$ and α can change the width of the confidence interval. But what about the sample size?
- Well we know that this changes the standard deviation of the sampling distribution... as n goes up, SD goes down!
 - if SD goes down, then width of confidence interval gets smaller
- Goal: We want to use this relationship to determine how large my sample size should be.

Margin of Error

- The **margin of error** represents the distance from $\hat{p}$ to one end of the interval.

- Recall we have $\left(\hat{p} - z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}}, \hat{p} + z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}} \right)$

- I can write the margin of error as $ME = z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}}$

Margin of Error

- The total length of the interval is 2 times the margin of error
- As mentioned previously, the sample size plays a roll in deciding how big the margin of error will be.
 - as does the confidence level, and the choice for p in the standard error, but we will focus now on the sample size
- Key Idea: If we know ahead of time that we want an interval with a certain width and at a specific confidence level, I can figure out how big of a sample I need.

Margin of Error

- I can write the margin of error as $ME = z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}}$
- If I decide how wide my interval should be (ME) and the confidence level (α) and thus $z_{\alpha/2}$, I can find a value for n, by rearranging the above expression.
- This would give me the expression $n = \left(\frac{z_{\alpha/2} \sqrt{p(1-p)}}{ME} \right)^2$
- So if I want to create a 95% confidence interval, where the total width of the interval is 0.2 (i.e. 0.1 on either side of $\hat{p}$), then I have $z_{\alpha/2} = z_{0.05/2} = 1.96$ and $ME = 0.1$.
- What to do with p?

Conservative choice of p

- This is where the idea of the conservative choice of p comes in.
- The reason why we want to use $p = 0.5$ to calculate a sample size is because we know this will give us the biggest possible standard error of our sampling distribution
 - and thus the widest interval.
- Using $p = 0.5$ means we will always get a sample size with more data than we should need to make a confidence interval with a particular width.
- And it is never a bad thing to have too much data!

Sample size formula

- So we should always use $p = 0.5$ to calculate a sample size because it will give us more data than we need.
- We can now write the final formula to use to find the sample size for proportion:

$$n = \left(\frac{z_{\alpha/2} \times 0.5}{ME} \right)^2$$

- Always round up to the nearest integer
- you are collecting units/objects so you can only collect whole units (not fractions)
- you also want to have enough, so that's why we round up and never down.

Example: Health Care

In a 2019 Canada Day poll of 1000 Canadians, 58% said they were proud of Canadian health care.

- A. Find the margin of error for the poll if we want 90% confidence in our estimate. Use a conservative choice for p .
- B. What does this margin of error mean?

$$A) \hat{p} = 0.58 \quad n = 1000$$

$$90\% \text{ Confidence} \Rightarrow \alpha = 0.1 \Rightarrow Z_{\alpha/2} = 1.645$$

$$ME = Z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}} = 1.645 \sqrt{\frac{0.5(1-0.5)}{1000}} = 0.026$$

B) In a 90% CI for p , the distance from the centre ($\hat{p}$) to the end of the interval is 0.026.

The total width of a 90% CI would be 2×0.026 .

Example: Health Care

C. If we want to be 99% confident, would the margin of error be larger or smaller?

D. If we want to be 99% confident but have a margin of error of 0.02, how many people should we survey?

c) Larger.

If we want to be more confident, we need a wider interval.

D) 99% confidence $\Rightarrow \alpha = 0.01 \Rightarrow Z_{\alpha/2} = 2.575$



$q_{\text{norm}}(0.005)$

$$n = \left(\frac{Z_{\alpha/2} \times 0.5}{ME} \right)^2 = \left(\frac{2.575 \times 0.5}{0.02} \right)^2 = 4144.14$$

which we round UP to 4145

Confidence Intervals for the mean

Confidence Intervals for Means

- So far, we have built confidence intervals around the population proportion p . This is appropriate when the population parameter we are interested in is a proportion.
- However, there are other cases when we are instead interested in the population mean μ .
- Recall that we are also able to use CLT to obtain the sampling distribution of the sample mean $\bar{X}$. We can use this to build a confidence interval for the population mean μ in a very similar way.

Sampling distribution of the mean

- Recall from the CLT that for a sufficiently large n ,

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

where μ is the population mean of the quantity measured and σ^2 is the population variance of the quantity measured.

- This can be standardized so that

$$\frac{\bar{X} - \mu}{\sqrt{\sigma^2/n}} \sim N(0,1)$$

- We can use this to construct a confidence interval just like before

Deriving the confidence interval

$\nearrow N(0,1)$

$$P(-1.96 < Z < 1.96) = 0.95$$

Sub in standardized $\bar{X}$

$$P\left(-1.96 < \frac{\bar{X} - \mu}{\sqrt{\sigma^2/n}} < 1.96\right) = 0.95$$

multiply by
denominator

$$P\left(-1.96\sqrt{\sigma^2/n} < \bar{X} - \mu < 1.96\sqrt{\sigma^2/n}\right) = 0.95$$

Subtract $\bar{X}$

$$P\left(-\bar{X} - 1.96\sqrt{\sigma^2/n} < -\mu < -\bar{X} + 1.96\sqrt{\sigma^2/n}\right) = 0.95$$

multiply by
-1

$$P\left(\bar{X} + 1.96\sqrt{\sigma^2/n} > \mu > \bar{X} - 1.96\sqrt{\sigma^2/n}\right) = 0.95$$

Rearrange

$$P\left(\bar{X} - 1.96\sqrt{\sigma^2/n} < \mu < \bar{X} + 1.96\sqrt{\sigma^2/n}\right) = 0.95$$

Therefore, the 95% confidence interval for μ is

$$\left(\bar{X} - 1.96\frac{\sigma}{\sqrt{n}}, \bar{X} + 1.96\frac{\sigma}{\sqrt{n}}\right)$$

Any level of confidence

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

- Similar to before, the level of confidence can be anything you want it to be.
- In general, for a $(1 - \alpha)\%$ confidence interval, we will need to find the critical value $z_{\alpha/2}$ such that $P(Z < -z_{\alpha/2}) = \alpha/2$.
- This gives a confidence interval of

$$\left(\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

- Also can be written as

$$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \quad \text{or} \quad \bar{X} \pm z_{\alpha/2} \times SE(\bar{X})$$

Unknown σ

- The formula is not in terms of μ which is good since that's the unknown population parameter that we're trying to learn about
- The trouble here is that, just like we don't know the true population mean or proportion, we also don't know what the **population SD σ** is.
- Similar to one of the options with CIs for proportions, we can use a value computed from our sample:
 - In this case, it makes sense to use the **sample standard deviation s**
 - This is the standard deviation that we can compute based on our data
 - We use $\frac{s}{\sqrt{n}}$ to estimate the standard deviation of the sampling distribution $\frac{\sigma}{\sqrt{n}}$

Sample Standard Deviation

- Recall from Week 1 when we were looking at summary statistics
- Sample Variance = $s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$
- Sample standard deviation = $s = \sqrt{s^2} = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}$
- In R, we can calculate the sample variance and sample standard deviation using the `var()` and `sd()` commands

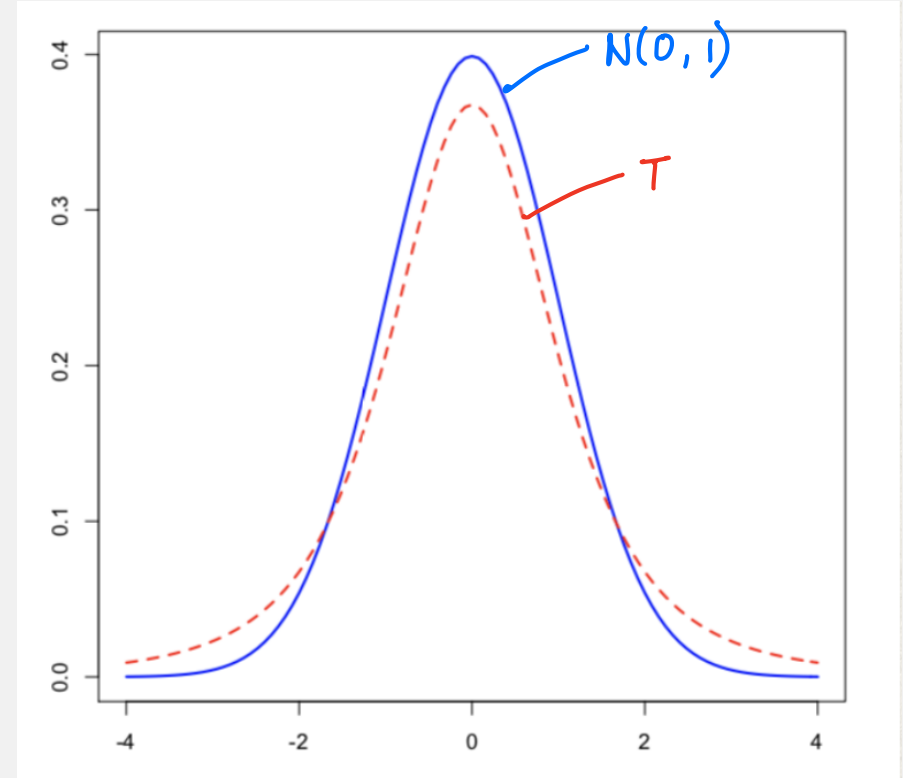
Extra Variation

- We encounter another snag though when working with means:
 - Our interval is now using two pieces of information that we are computing from our sample data: $\bar{x}$ and s
 - Every time we estimate something, we are introducing more uncertainty/variability
 - Our formula for the CI of the mean only accounts for the variability of using $\bar{x}$ to estimate μ . It doesn't account for the extra variability of using s to estimate σ
 - Then, even though we say we are building a $(1 - \alpha)\%$ interval, that may not be what we actually get!
 - So our interval may not actually have the **coverage** that we want.
- Solution: Instead of using $N(0,1)$ to find the critical values, we will need to use another distribution.

T Distribution

T Distribution

- The T distribution is a probability distribution that is also symmetric, unimodal, and bell-shaped
- It is always centered at 0
- The main difference from $N(0,1)$ is the probability of getting values far from the mean (i.e. in the tails of the distribution)
- The T distribution has heavier tails, so bigger chance of being far from the mean.

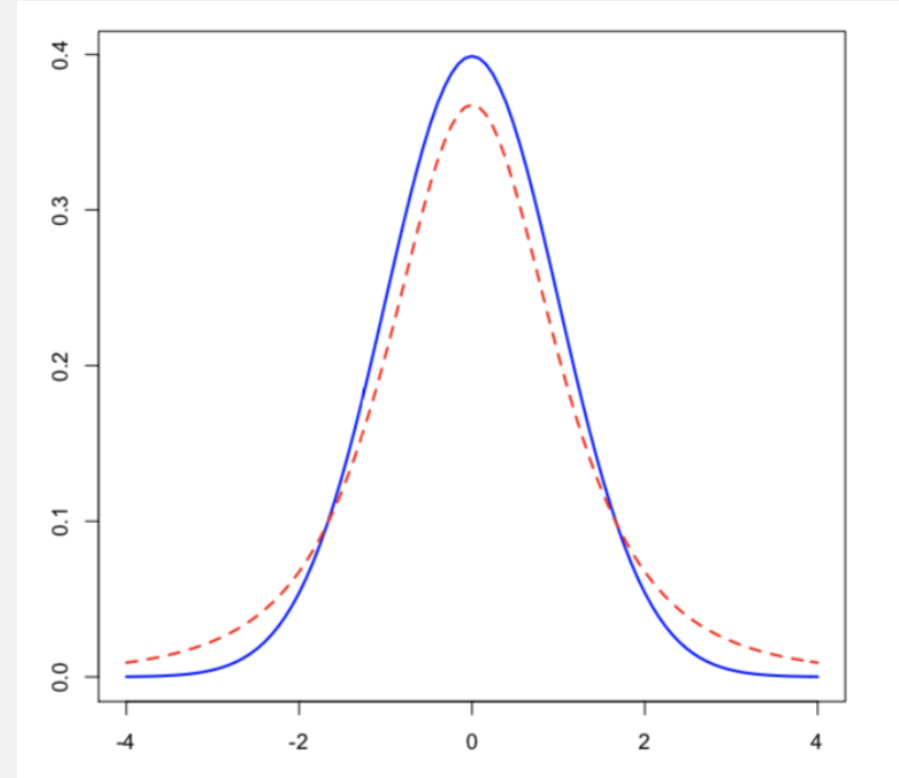


T Distribution

T_2

$T_{df=100}$

- The T distribution only has one parameter that controls its shape, called the **degrees of freedom (df)**
- This is written as $X \sim T_{df}$
- To fully define the T distribution, you need to specify the df
- Turns out that as the df gets larger and larger, the T distribution more closely resembles $N(0,1)$

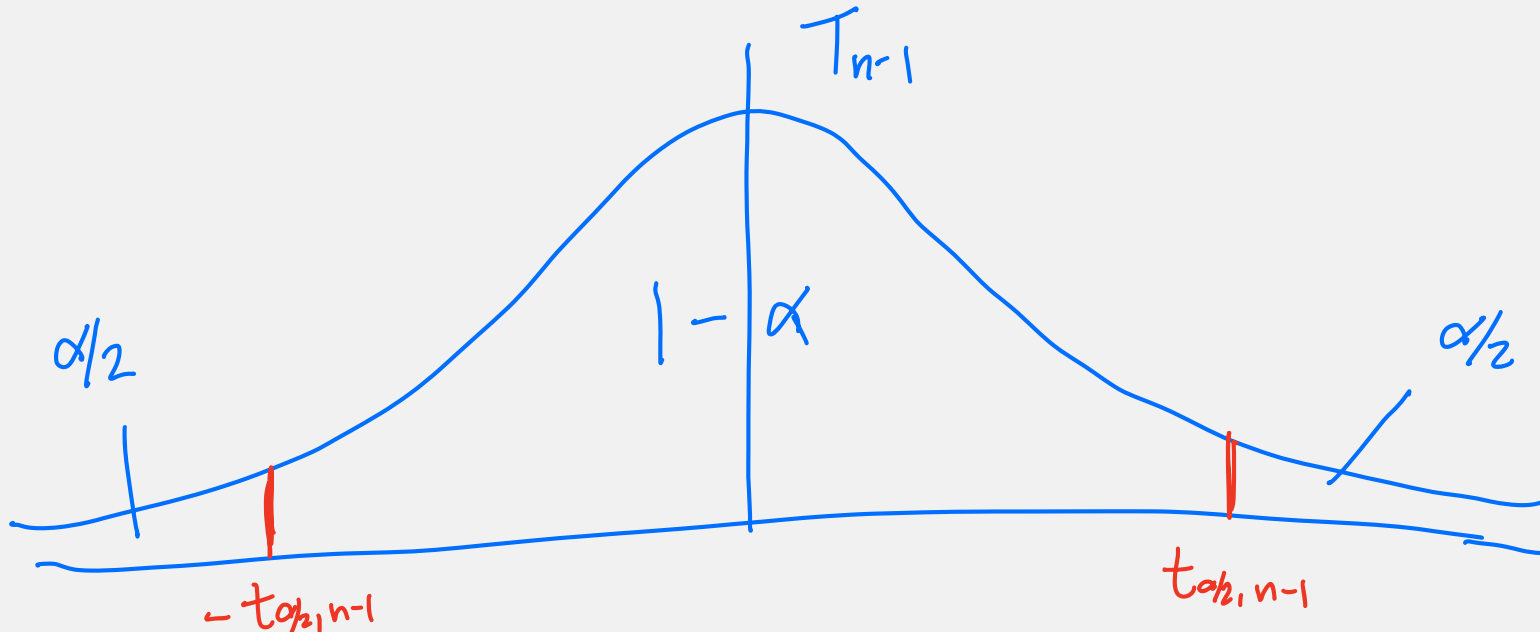


Unknown σ

- So why does this help us when we don't know σ ?
 - If we need to estimate σ with s , we need to account for the fact that both our estimates for σ and μ will vary with each different sample we take.
 - more variability in two places means higher chance of getting extreme sample values
 - To have the right coverage for the interval, we need to use a distribution that also has higher chance of getting extreme values
- In order to find the critical values, we will want to use the T distribution with $df = n - 1$ where n is the sample size

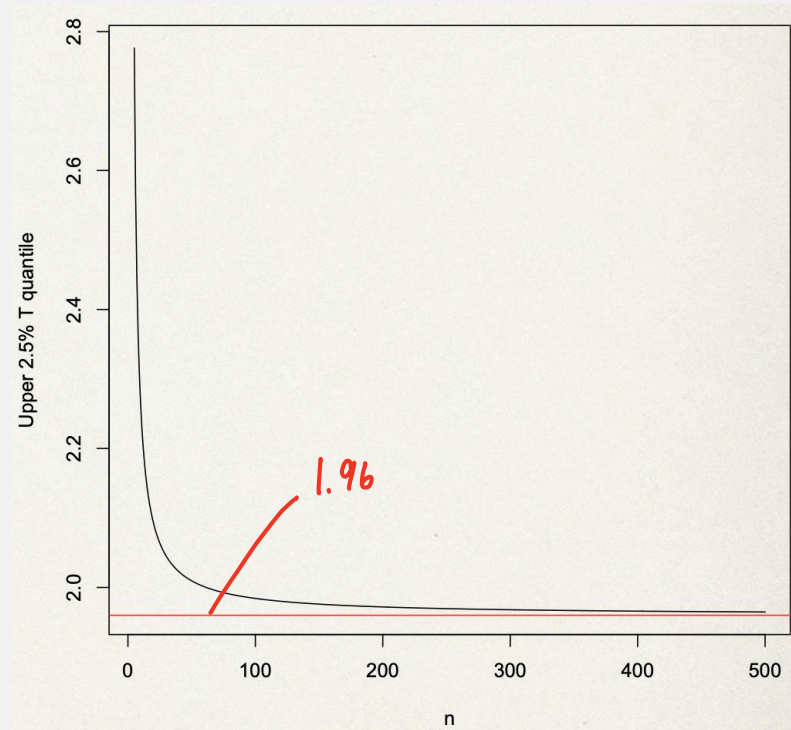
Finding the critical value

- Finding the critical value is the same as before
- We are interested in a value $t_{\frac{\alpha}{2}, n-1}$ such that $P(T_{n-1} < -t_{\frac{\alpha}{2}, n-1}) = \frac{\alpha}{2}$



Finding the critical value for 95% CI

- Say we want to build a 95% confidence interval around the mean.
- We then need quantiles for $\alpha = 0.05$
- Since the T distribution parameter is based on the sample size n , we can look at how larger sample sizes get us closer to a $N(0, 1)$.
 - we know $z_{\alpha/2} = 1.96$ (red line)
 - by the time we have a sample of size of 100, the T distribution quantile for the same $t_{df=99}$ is already 1.98
 - So, very close to the Normal one

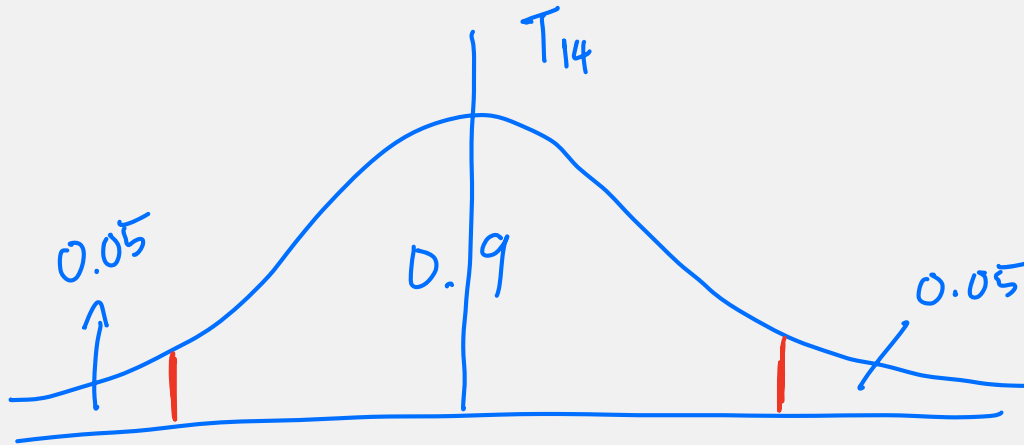


T distribution in R

- Similar to how we use `pnorm()` and `qnorm()` for the Normal distribution, we have corresponding R functions for the T distribution
- Use `pt(q, df=?)` to calculate $P(T_{df=?} < q)$
- Use `qt(p, df=?)` to find q such that $P(T_{df=?} < q) = p$

Example

Find the critical value for a 90% CI using the T-distribution where $n = 15$



$$qt(0.05, df = 14) \rightarrow -1.76$$

T Distribution and CIs

- By using the T distribution, especially when sample size is small(ish), we can account for the fact that $\bar{x}$ and s may not be good estimates
 - use a heavier tailed distribution to see higher chance of extreme values
 - the T distribution will make our CI wider because the quantile is bigger than the Normal
- Just like sample proportions and means, s gets better (and closer to σ) when sample size goes up.
 - then T distribution gets closer to Normal, and we no longer need to worry about extreme estimates

Confidence Intervals for Means, Unknown σ

- Now that we understand why we need to use the T distribution when dealing with means and their confidence intervals, we can look at the interval itself.
- Just like with proportions, we can specify whatever confidence level $(1 - \alpha)\%$ we want
 - For some α , find the corresponding quantile of the T distribution, $t_{\frac{\alpha}{2}, n-1}$ to be the critical value
 - Estimate both $\bar{x}$ and $s = \sqrt{\frac{\sum_i (x_i - \bar{x})^2}{n-1}}$ from your sample
 - Compute the $(1 - \alpha)\%$ CI: $\left(\bar{X} - t_{\frac{\alpha}{2}, n-1} \times \frac{s}{\sqrt{n}}, \bar{X} + t_{\frac{\alpha}{2}, n-1} \times \frac{s}{\sqrt{n}} \right)$
 - In rare cases where σ is known, you can use the CI based on the Normal distribution

Sample Sizes

Size of a Confidence Interval

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}}$$

- Similar to before, the margin of error is the distance between the sample mean and the end of the confidence interval
- $ME = \text{critical value} \times \frac{\sigma}{\sqrt{n}}$ or $ME = \text{critical value} \times \frac{s}{\sqrt{n}}$
- General format of the CI is: sample estimate $\pm$ margin of error
- We can see the width of the CI is impacted by several things:
 - The confidence level $(1 - \alpha)$
 - Higher confidence means wider CI
 - Standard deviation σ or sample standard deviation s
 - Higher SD means wider CI
 - Sample size n
 - Larger sample size means narrower CI

Calculating the required sample size

- We can re-arrange the formula for the margin of error to get the following:

- If σ is known, then $n = \left(\frac{z_{\alpha/2} \times \sigma}{ME} \right)^2$

- If σ is NOT known, then $n = \left(\frac{t_{\alpha/2, n-1} \times s}{ME} \right)^2$

- But we have a problem here! The formula requires a critical value with $df=n-1$. Which means that the critical value depends on n .

- So, let's just use the normal distribution

- Therefore, you can calculate the required sample size by using $n = \left(\frac{z_{\alpha/2} \times s}{ME} \right)^2$ and then rounding up!

Summary

Summary of Notation

- Population Parameters: Unknown characteristics of whole population
 - Population proportion: p
 - Population mean: μ
 - Population standard deviation: σ
 - Population variance: σ^2
- Sample Statistics: Summary measures of sample data
 - Sample proportion: $\hat{p}$
 - Sample mean: $\bar{X} = \hat{\mu}$
 - Note: $\bar{X}$ refers to the sample mean as a random variable, $\bar{x}$ refers to a computed value based on a specific dataset
 - Sample standard deviation: $s = \hat{\sigma}$
 - Sample variance: $s^2 = \hat{\sigma}^2$

Summary of CI for p

- A $(1 - \alpha)\%$ CI can be expressed as:

$$\hat{p} \pm z_{\alpha/2} \times \sqrt{\frac{p(1 - p)}{n}}$$

- Use either 0.5 or $\hat{p}$ in place of p in the formula

Summary of CI for μ

- When σ is known, a $(1 - \alpha)\%$ CI can be expressed as:

$$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

- When σ is NOT known, a $(1 - \alpha)\%$ CI can be expressed as:

$$\bar{X} \pm t_{\frac{\alpha}{2}, n-1} \times \frac{s}{\sqrt{n}}$$

Example: Parking Fees

A city builds a new parking structure in the central business district. The city plans to pay for the structure through parking fees. During a 44 day period, the daily parking fees collected were on average \$126 with a standard deviation of \$15.

- a) Find a 90% confidence interval for the mean daily income this parking structure will generate.
- b) The consultant who advised the city thinks the parking structure fees will generate an average of \$130 of revenue for the city. Based on your interval, do you think his guess is correct?
- c) Recalculate the 90% confidence interval with the additional information that the true standard deviation of parking fees collected is \$15.
- d) Compare the answers in part a and c. Do they make sense?

a) We are interested in μ , which is the average parking fees collected in a day.

$$n = 44 \quad \bar{x} = 126 \quad s = 15$$

$$t_{\alpha/2, n-1} = 1.68 \quad \text{from} \quad qt(p=0.05, df=43)$$

$$\begin{aligned} 90\% \text{ CI} : \quad \bar{x} \pm t_{\alpha/2, n-1} \times \frac{s}{\sqrt{n}} \\ \Rightarrow 126 \pm 1.68 \times \frac{15}{\sqrt{44}} \\ \Rightarrow (122.20, 129.80) \end{aligned}$$

We are 90% confident that the daily avg parking fees is between \$122.20 and \$129.80.

b) No, based on the 90% Confidence interval, it does not support the claim that the city will make \$130 daily.

$$c) \quad \sigma = 15 \quad n = 44 \quad \bar{x} = 126$$

$$Z_{\alpha/2} = 1.64$$

$$\begin{aligned} 90\% \text{ CI} : \quad \bar{x} \pm Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}} \\ \Rightarrow 126 \pm 1.64 \times \frac{15}{\sqrt{n}} \\ \Rightarrow (122.29, 129.71) \end{aligned}$$

d) Yes. interval is a bit wider when σ is unknown.